Chapter 14: Sleep Neuroscience

Comprehensive Neuroscience Reference

Abstract

Sleep is a reversible state of reduced consciousness and motor activity occupying roughly one-third of human life. This chapter examines the neurobiological mechanisms underlying sleep architecture, homeostatic and circadian regulation, memory consolidation, and sleep disorders. Sleep is organized into recurring ∼90-minute cycles comprising NREM stages (N1 light; N2 with sleep spindles and K-complexes; N3 deep slow-wave sleep with high-amplitude delta oscillations) and REM sleep (desynchronized EEG, rapid eye movements, muscle atonia, vivid dreaming). The two-process model explains sleep-wake timing: homeostatic Process S (adenosine accumulation during wakefulness, driving sleep pressure) is modulated by circadian Process C (SCN-generated daily alerting rhythm). A bistable flip-flop switch between sleep-promoting and wake-promoting circuits (stabilized by orexin) creates sharp state transitions. During sleep, coordinated slow oscillations, spindles, and hippocampal sharp-wave ripples mediate memory consolidation (systems consolidation of declarative memories). The glymphatic system — a brain-wide waste clearance pathway — is most active during sleep, clearing amyloid-beta and other metabolites; sleep deprivation impairs this clearance, contributing to neurodegeneration risk. Sleep disorders (insomnia, obstructive sleep apnea, narcolepsy, REM behavior disorder) cause daytime somnolence, cognitive impairment, and neurodegeneration via multiple mechanisms (sleep fragmentation, intermittent hypoxia, loss of memory consolidation, impaired glymphatic clearance).

Contents

1 Sleep Architecture **2**
1.1 NREM (Non-Rapid Eye Movement) Sleep . . . 2
1.2 REM (Rapid Eye Movement) Sleep (∼20–25% of total sleep) . . . 2

2 Sleep-Wake Regulation: The Two-Process Model **3**
2.1 Process S (Homeostatic Sleep Pressure) . . . 3
2.2 Process C (Circadian Alerting Signal) . . . 3

3 Neural Circuits of Sleep and Wakefulness **3**
3.1 Wake-Promoting Systems . . . 4
3.2 Sleep-Promoting Systems . . . 4
3.3 The Flip-Flop Switch (Saper, 2001) . . . 4
3.4 REM Sleep Circuitry . . . 4

4 The Glymphatic System **4**
4.1 Mechanism . . . 5

4.2 Sleep Enhancement 5
4.3 Clinical Significance 5

5 Functions of Sleep **5**
5.1 Memory Consolidation 5
5.2 Synaptic Homeostasis Hypothesis (SHY) — Tononi & Cirelli 5
5.3 Immune Function 6
5.4 Emotional Regulation 6

6 Sleep Disorders **6**
6.1 Insomnia 6
6.2 Obstructive Sleep Apnea (OSA) 6
6.3 Narcolepsy 7
6.4 REM Sleep Behavior Disorder (RBD) 7

References **7**

1. Sleep Architecture

Humans spend approximately one-third of their lives asleep. Sleep is organized into recurring cycles (∼90 minutes each; 4–6 cycles per night) comprising two fundamentally distinct states:

1.1 NREM (Non-Rapid Eye Movement) Sleep

Stage N1 (light sleep, ∼5% of total sleep):

- Transition from wakefulness; easily aroused
- EEG: Theta waves (4–8 Hz), vertex sharp waves
- Slow rolling eye movements; decreased muscle tone
- Hypnagogic hallucinations, hypnic jerks (myoclonic twitches)

Stage N2 (intermediate sleep, ∼50% of total sleep):

- **Sleep spindles** (12–16 Hz, 0.5–2 s bursts): Generated by thalamic reticular nucleus → thalamocortical relay neurons → cortex. Functions: protect sleep from external disturbances (sensory gating), promote synaptic plasticity and memory consolidation (Ca^{2+} influx during spindle oscillations → CaMKII activation → LTP). Spindle density correlates with IQ measures and learning ability.
- **K-complexes** (large biphasic waveforms): Evoked by external stimuli or generated spontaneously. Reflect cortical down-states; may serve a sleep-protective function (suppressing arousal) and contribute to slow-wave activity buildup.

Stage N3 (deep/slow-wave sleep, ∼20–25% of total sleep):

- EEG: High-amplitude delta waves (0.5–4 Hz) and slow oscillations (<1 Hz)
- **Slow oscillations** (∼0.75 Hz): Consist of alternating UP states (neuronal depolarization and firing) and DOWN states (hyperpolarization and silence). Generated by cortical Layer V pyramidal neurons; propagate as traveling waves across the cortex. UP states provide temporal windows for hippocampal sharp-wave ripples and thalamocortical spindles — the three-way coordination critical for memory consolidation.
- Highest arousal threshold; growth hormone release peak
- Predominates in the first half of the night; decreases with age

1.2 REM (Rapid Eye Movement) Sleep (∼20–25% of total sleep)

- EEG: Desynchronized, low-amplitude, mixed-frequency (similar to wakefulness); prominent **theta rhythm** in hippocampus; **PGO waves** (pontine-geniculate-occipital — phasic bursts)
- **Rapid eye movements**: Conjugate saccade-like movements, generated by pontine burst neurons
- **Muscle atonia**: Active inhibition of spinal motor neurons via glycinergic/GABAergic circuits originating in the sublaterodorsal nucleus (SLD/SubC) → ventromedial medulla → spinal motor neurons. Loss of REM atonia → REM sleep behavior disorder (acting

out dreams — a strong prodromal marker for α-synucleinopathies: Parkinson's, Lewy body dementia)

- **Dreaming**: Most vivid and narrative-rich dreams occur during REM
- **Autonomic instability**: Variable heart rate, blood pressure, and breathing (irregular)
- **Sexual arousal**: Penile erections (males), clitoral engorgement (females) — independent of dream content
- Predominates in the second half of the night

2. Sleep-Wake Regulation: The Two-Process Model

Sleep timing is governed by two independent processes (Borbély, 1982):

2.1 Process S (Homeostatic Sleep Pressure)

Sleep pressure builds exponentially during wakefulness and dissipates exponentially during sleep (primarily during NREM slow-wave sleep).

Molecular basis: **Adenosine** accumulation in the basal forebrain and cortex during wakefulness. Adenosine is a byproduct of ATP metabolism; its concentration increases with sustained neural activity. Adenosine acts on:

- **A1 receptors** (inhibitory): Suppresses wake-promoting neurons in basal forebrain and cortex → promotes sleepiness
- **A2A receptors** (excitatory): Activates sleep-promoting neurons in the ventrolateral preoptic area (VLPO) and nucleus accumbens

Caffeine blocks A1 and A2A receptors → counteracts sleep pressure (but does not reduce the actual homeostatic need for sleep — sleep debt accumulates).

Other homeostatic factors: Prostaglandin D_2, cytokines (IL-1β, TNF-α — link between infection/inflammation and sleepiness), GABA accumulation, and astrocytic adenosine release.

2.2 Process C (Circadian Alerting Signal)

The SCN generates a circadian rhythm of alerting that opposes sleep pressure during the day. This signal strengthens throughout the day, peaking in the evening ("wake maintenance zone" — the hardest time to fall asleep, ∼2–3 hours before habitual bedtime, despite high sleep pressure). The circadian alerting signal drops sharply at night, coinciding with melatonin onset and the opening of the "sleep gate."

The two processes interact: sleep occurs when Process S exceeds Process C threshold; waking occurs when Process S has dissipated sufficiently. Jet lag and shift work create misalignment between S and C.

3. Neural Circuits of Sleep and Wakefulness

3.1 Wake-Promoting Systems

Key wake-promoting nuclei: Locus coeruleus (NE), raphe nuclei (5-HT), tuberomammillary nucleus (histamine), basal forebrain (ACh), lateral hypothalamus (orexin/hypocretin), ventral periaqueductal gray (DA), pedunculopontine/laterodorsal tegmental nuclei (ACh — also REM-on).

3.2 Sleep-Promoting Systems

Ventrolateral preoptic area (VLPO): The primary sleep-promoting nucleus. Contains GABAergic and galaninergic neurons that inhibit all major wake-promoting nuclei during sleep. VLPO neurons are active during NREM sleep; damage → severe insomnia (as in fatal familial insomnia, though this disease primarily involves thalamic degeneration).

Median preoptic nucleus (MnPO): Also contains sleep-active GABAergic neurons; may respond to homeostatic sleep pressure and initiate sleep onset.

Parafacial zone (PZ): GABAergic neurons in the medullary brainstem; promote NREM slow-wave sleep by inhibiting the parabrachial wake-promoting nucleus.

3.3 The Flip-Flop Switch (Saper, 2001)

VLPO and wake-promoting nuclei **mutually inhibit** each other, creating a bistable switch with two stable states (awake or asleep) and rapid transitions between them. **Orexin/hypocretin neurons** stabilize the switch in the wake position by exciting wake-promoting nuclei. Loss of orexin → unstable switch → inappropriate state transitions → **narcolepsy type 1** (excessive daytime sleepiness + cataplexy — sudden loss of muscle tone triggered by emotions).

3.4 REM Sleep Circuitry

REM-on neurons: Sublaterodorsal nucleus (SLD/SubC; glutamatergic) — generates REM sleep. Projects to:

- Ventromedial medulla → spinal motor neuron inhibition → atonia
- Basal forebrain → cortical activation
- PPT/LDT → cholinergic activation of cortex

REM-off neurons: Ventrolateral periaqueductal gray (vlPAG) and lateral pontine tegmentum (LPT) — GABAergic; inhibit SLD during wakefulness and NREM.

REM flip-flop: Mutual inhibition between REM-on (SLD) and REM-off (vlPAG/LPT) neurons creates a second bistable switch within sleep, controlling NREM↔REM transitions. Orexin stabilizes this switch in the REM-off position (explaining why narcolepsy includes frequent inappropriate REM intrusions).

4. The Glymphatic System

Discovered by Nedergaard and colleagues (2012), the glymphatic system is a brain-wide waste clearance pathway that is most active during sleep.

4.1 Mechanism

1. **CSF enters brain**: Arterial pulsations drive CSF from the subarachnoid space along perivascular (Virchow-Robin) spaces surrounding penetrating arterioles
2. **CSF enters interstitial space**: Passes through **aquaporin-4 (AQP4) water channels** on astrocyte endfeet lining the perivascular space
3. **Convective flow**: CSF mixes with interstitial fluid (ISF), flushing solutes through the brain parenchyma
4. **Waste collection**: ISF and solutes collect along perivenous channels and drain into cervical lymphatic vessels and meningeal lymphatic vessels (rediscovered in 2015 — overturned the dogma that the brain lacked lymphatics)

4.2 Sleep Enhancement

Glymphatic clearance is **∼60% more efficient during sleep** compared to wakefulness (and during anesthesia). The mechanism: during sleep, **interstitial space volume increases ∼60%** (neurons shrink) due to decreased noradrenergic tone from the locus coeruleus. This expansion reduces resistance to convective flow.

4.3 Clinical Significance

- **Amyloid-β clearance**: Aβ (the amyloid precursor protein fragment that accumulates in Alzheimer's disease) is cleared via the glymphatic system. Sleep deprivation → impaired Aβ clearance → increased Aβ accumulation. This may explain the epidemiological link between chronic sleep disruption and increased Alzheimer's risk.
- **Tau clearance**: Tau protein spread may also be facilitated by disrupted glymphatic flow
- **Traumatic brain injury**: TBI impairs glymphatic function → reduced waste clearance → may contribute to post-traumatic neurodegeneration (CTE)
- **Aging**: Glymphatic function declines with age (AQP4 depolarization from endfeet, arterial stiffening → reduced pulsatile drive) — potentially contributing to age-related protein aggregation disorders

5. Functions of Sleep

5.1 Memory Consolidation

NREM slow-wave sleep: Hippocampal-to-cortical transfer of declarative memories via coordinated slow oscillations, spindles, and sharp-wave ripples.

REM sleep: Procedural memory consolidation, emotional memory processing, creative insight.

5.2 Synaptic Homeostasis Hypothesis (SHY) — Tononi & Cirelli

Core idea: Wakefulness is associated with net synaptic potentiation (learning strengthens synapses throughout the day). This increases energy demands, space requirements, and signal-to-noise ratio. NREM slow-wave sleep provides **global synaptic downscaling** — a propor-

tional reduction in synaptic strength that renews learning capacity, reduces energy costs, and improves signal-to-noise.

Evidence: Synaptic markers (AMPA receptor phosphorylation, spine size, slope of cortical evoked potentials) increase with wakefulness and decrease with sleep. Sleep deprivation saturates LTP capacity; recovery sleep restores it.

5.3 Immune Function

Sleep enhances immune function:

- NREM sleep promotes pro-inflammatory cytokine production (IL-1β, TNF-α) and T-cell redistribution to lymph nodes
- Sleep deprivation impairs antibody response to vaccination, reduces natural killer cell activity, and increases susceptibility to infection (sleeping <6 hours $\rightarrow$ 4.2$\times$ increased risk of catching a cold)
- Bidirectional: Infection $\rightarrow$ increased sleep (via cytokine-mediated activation of sleep circuits)

5.4 Emotional Regulation

REM sleep recalibrates emotional reactivity:

- Overnight REM sleep reduces amygdala response to previously encountered emotional stimuli
- The neurochemical milieu of REM (absent norepinephrine from LC; high acetylcholine) may allow emotional memories to be reprocessed without the arousal signature — "sleeping off" emotional distress
- Sleep deprivation $\rightarrow$ $\sim$60% increase in amygdala reactivity to negative stimuli, with reduced amygdala-mPFC functional connectivity (impaired top-down regulation)

6. Sleep Disorders

6.1 Insomnia

Most prevalent sleep disorder ($\sim$10% chronic). Characterized by difficulty initiating or maintaining sleep despite adequate opportunity. Neurobiological model: Hyperarousal — elevated cortisol, increased whole-brain glucose metabolism during NREM sleep (insufficient deactivation of wake circuits), smaller VLPO cell populations. Treatment: Cognitive behavioral therapy for insomnia (CBT-I, first-line), pharmacotherapy (GABA-A modulators, orexin receptor antagonists, melatonin receptor agonists).

6.2 Obstructive Sleep Apnea (OSA)

Repetitive upper airway collapse during sleep $\rightarrow$ intermittent hypoxia + sleep fragmentation. Prevalence: $\sim$10–30% of adults. Consequences: excessive daytime sleepiness, hypertension, cardiovascular disease, cognitive impairment (particularly executive function and memory —

hippocampal and PFC gray matter reduction), and increased risk of neurodegenerative disease. Treatment: Continuous positive airway pressure (CPAP).

6.3 Narcolepsy

Type 1: Loss of orexin/hypocretin neurons (autoimmune destruction; strong HLA-DQB1*06:02 association) → excessive daytime sleepiness, cataplexy, sleep paralysis, hypnagogic hallucinations, fragmented nighttime sleep. CSF orexin-A levels <110 pg/mL (vs. >200 pg/mL normal) is diagnostic.

Type 2: Similar symptoms without cataplexy; orexin levels usually normal; mechanism unclear.

6.4 REM Sleep Behavior Disorder (RBD)

Loss of normal REM atonia → patients physically act out dreams (often violent or action-filled). >80% of idiopathic RBD patients eventually develop an α-synucleinopathy (Parkinson's disease, Lewy body dementia, or multiple system atrophy) — RBD is the strongest known prodromal marker, often preceding motor symptoms by 10–15 years. Mechanism: Neurodegeneration of brainstem REM atonia circuits (SLD → ventromedial medulla pathway).

References

1. Borbély, A. A. (1982). A two process model of sleep regulation. *Human Neurobiology*, 1(3), 195–204.
2. Fuller, P. M., Gooley, J. J., & Saper, C. B. (2006). Neurobiology of the sleep-wake cycle: Sleep architecture, circadian regulation, and regulatory mechanisms. *Journal of Biological Rhythms*, 21(6), 482–493.
3. Mignot, E. (2008). Why we sleep: The temporal organization of consciousness. *Oxford University Press*.
4. Nedergaard, M., & Goldman, S. A. (2016). Glymphatic failure as a final common pathway to dementia. *Alzheimers & Dementia*, 12(12), 1107–1116.
5. Saper, C. B., & Fuller, P. M. (2017). Wake–sleep circuitry: An evolving story. *Neuroscience*, 338, 186–193.
6. Scammell, T. E., Arrigoni, E., & Lipton, J. O. (2017). Neural circuitry of wakefulness and sleep. *Neuron*, 93(4), 747–765.
7. Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: A hypothesis. *Neuron*, 75(1), 15–35.
8. Xie, L., Kang, H., Xu, Q., et al. (2013). Sleep drives metabolite clearance from the adult brain. *Science*, 342(6156), 373–377.